LEARNING SCIENCES LAB · STUDY BRIEF

Five-minute quizzes beat re-reading — by a full letter grade

A classroom randomized trial of spaced retrieval across 1,800 students.

Short daily quizzes raised final-exam scores by 11 points over re-reading.

We ran a year-long randomized trial in introductory biology. Students who ended each class with a five-minute low-stakes quiz scored 11 percentage points higher on the cumulative final than students given the same five minutes to re-read. The effect held across instructors.

The gap we set out to close: lab effects rarely survive a real classroom.

Retrieval practice is one of the best-replicated findings in cognitive psychology — in the lab. But labs use word lists and college sophomores under controlled conditions. Whether the effect survives a messy semester, real content, and ordinary teachers was the open question.

- Most prior evidence comes from short laboratory sessions.
- Classroom replications are scarce and underpowered.

Design: 1,800 students, 24 sections, randomized within instructor.

Across four campuses, 24 sections of introductory biology were randomized at the section level to retrieval practice or a re-reading control. Both spent the same five minutes on the same material. We blinded the graders and pre-registered the analysis.

- Section-level randomization, balanced within each instructor.
- Pre-registered primary outcome: cumulative final-exam score.

What each condition actually did in the last five minutes of class.

Retrieval sections answered three short free-recall questions, then saw the answers — no grade attached. Control sections re-read the same three passages. Identical content, identical time, one difference: producing the answer versus reviewing it.

- Retrieval: recall first, feedback second.
- Control: re-read the same passage, no recall.

PRIMARY OUTCOME · CUMULATIVE FINAL EXAM

Retrieval practice moved the whole distribution, not just the top.

n = 1,800, intention-to-treat, clustered standard errors by section.

+11 pts
FINAL-EXAM GAIN

$d = 0.54$
EFFECT SIZE

1,800
STUDENTS ENROLLED

$p<0.001$
VS RE-READING

The benefit was largest for the students who started behind.

When we split by incoming GPA, the lowest-performing quartile gained 16 points; the top quartile gained 6. Retrieval practice did not just lift the average — it narrowed the gap. The intervention was most useful where it was most needed.

- Bottom quartile: +16 points.
- Top quartile: +6 points.

Durability: the advantage was larger at the final than at the unit test.

On the weekly unit tests the retrieval gap was 7 points; on the cumulative final months later it was 11. The effect did not fade — it grew. Retrieval practice does not just help students learn; it helps them keep what they learned.

- Unit-test gap: +7 points.
- Final-exam gap: +11 points — the effect compounds.

Why it works: producing an answer is a different act than recognizing one.

Re-reading creates fluency — the comfortable sense that you know the material. Retrieval forces reconstruction, which strengthens the memory trace and, crucially, reveals what you do not yet know. The five minutes are spent diagnosing, not just reviewing.

- Recall strengthens the trace; re-reading only refreshes it.
- The feedback surfaces gaps the student cannot otherwise see.

Limits: it is one subject, and the quizzes were ungraded.

We tested introductory biology; transfer to writing-heavy or skills-based courses is untested. And the no-stakes design matters — adding grades may introduce anxiety that erodes the benefit. We report a robust effect within a specific, replicable setup.

- Single discipline; broader transfer unverified.
- Low-stakes framing may be load-bearing.

LEARNING SCIENCES LAB · PREPRINT AND MATERIALS AT RETRIEVALSTUDY.EXAMPLE

The cheapest intervention we tested was also the most effective.